

Cross-Attention for AES Mode Variation in Side-Channel Analysis

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Abstract—Portability poses a significant challenge for Deep Learning (DL)-based profiling Side-Channel Analysis (SCA) on AES encryption, as attackers cannot always ensure that training and target samples use the same encryption mode. To address this, we propose an Unsupervised Domain Adaptation (UDA) DL-SCA framework for achieving effective and robust cross-encryption-mode attacks. By incorporating cross-attention and UDA techniques, our framework aligns high-dimensional input samples, reducing interference from encryption mode mismatches. Evaluation across five distinct AES modes demonstrates that our method achieves robust SCA performance without requiring prior knowledge or multiple labeled datasets for analysis.

Index Terms—side channel analysis, cross-attention mechanisms, unsupervised domain adaptation, deep learning.

I. INTRODUCTION

Since Kocher's pioneering 1996 study on timing attacks [1], Side-Channel Analysis (SCA) has emerged as a critical technique for extracting sensitive information through the analysis of physical leakages [2]–[4]. One of the most powerful type of SCA, namely Profiling SCA, works by modeling leakages from similar devices to extract data from target devices [5], [6]. This approach was pioneered by Template Attacks (TA) [5], [7] and later enhanced by machine learning methods, such as Support Vector Machines (SVM) [8] and Random Forests (RF) [9]. The introduction of Deep Learning (DL) has further advanced profiling SCA. Specifically, deeper models such as Convolutional Neural Networks (CNN) and Multi-Layer Perceptrons (MLP) are capable of handling complex inputs more efficiently, allowing SCAs powered by such techniques to provide better results with regard to both the overall generalization capability and the performance in managing masked data [10]–[12].

In a DL-based SCA (henceforth DLSCA) against the AES encryption algorithm, the attacker's goal is to infer the secret key by correlating side-channel measurements from the encryption process with a leakage model (trained in the off-line reconnaissance phase with data collected from attacker-controlled devices) [4], [13]. Intuitively, this attack strategy is more likely to succeed if training samples of the leakage model is collected from the exact same hardware/software configure with the targeted devices, and we have indeed witnessed extensive DLSCA studies which only considered

the scenario where both the reconnoitered devices and the targeted ones run AES in the same encryption mode [14]–[16]. Unfortunately, practical attack scenarios could be more generalized, involving target devices that are different from the reconnoitered one at hardware-level, or alternatively, operate with AES modes different from the one observed during the off-line reconnaissance.

Related Work. Attentions on the portability of DLSCA, as well as the mode-specific AES vulnerabilities, has emerged only recently. While Wang *et al.* [17] analyzed the impact of mode diversity on DLSCA, Perusheska *et al.* [18] further highlighted the effectiveness of DLSCA using plaintext-ciphertext analysis across multiple AES modes. Other recent advancements focused on the scenario of across-device DLSCAs. Das *et al.* [19] demonstrated that training on diverse hardware in X-DeepSCA analysis enhances cross-device performance. Golde *et al.* [20] applied preprocessing methods, including dynamic time warping and Principal Component Analysis (PCA), with a MLP to mitigate cross-device discrepancies. Bhasin *et al.* [21] addressed portability challenges with a Multi-Device Model (MDM) that leverages three devices for training, validation, and attack. Cao *et al.* employed adversarial learning [22] and domain adaptation [23] to counteract device-based performance degradation, while Yu *et al.* [24] introduced meta-transfer techniques to address leakage inconsistencies in DLSCA across different device types. Nevertheless, researches on DLSCA that generalizes across multiple AES encryption modes remain limited. This gap underscores the need to study the possibility of building robust leakage models adaptable to diverse AES configurations, which could be the last piece of the puzzle on the completeness of DLSCA.

Contributions. Motivated by the current state-of-the-arts, we propose CA-SCA (Cross-Attention Side-Channel Analysis), a novel approach aiming to address the challenges in launching DLSCA against AES algorithms with the presence of mode variations. At the model design level, CA-SCA integrates cross-attention mechanisms with unsupervised domain adaptation (UDA) in order to have different sets of side-channel samples (which are potentially of different distributions) aligned at high-dimensional space to subsequently enable interactive training. To the best of our knowledge, while cross-attention has been widely used in multimodal [25] and

reinforcement learning [26] tasks [27], its application for AES mode adaptation in the SCA community remains unexplored. Thus our effort could be the first to make such an attempt.

In summary, the main contributions of this work are:

- We propose CA-SCA (Cross-Attention Side-Channel Analysis), a novel SCA approach that leverages cross-attention mechanisms and unsupervised domain adaptation to mitigate interference from mismatching of encryption modes, while requiring no specific preprocessing, fine-tuning, labeled attack traces or multiple analysis datasets.
- We evaluated CA-SCA on a software implementation of AES-128 across five encryption modes (ECB, CBC, CTR, CFB, and OFB) using the STM32F3 chip. The results demonstrate that CA-SCA reliably recovers keys across different encryption modes within 500 attack traces, outperforming current state-of-the-art methods¹.
- We present a comprehensive comparison of existing theoretical and experimental results on portable attacks using CA-SCA. Experimental results indicate that X-Deep and MDM may fail when encryption mode differences are substantial. In contrast, CA-SCA, using only a single mode as the analysis dataset, enables previously infeasible attacks.

The paper is organized as follows: Section II provides background knowledge and introduces the evaluation metrics. Section III outlines the proposed methodology. Section IV describes the experimental setup and presents the attack results, while Section V analyzes the impact of two key parameters on CA-SCA. Finally, Section VI concludes the paper and suggests future research directions.

II. BACKGROUND

A. DLSCA

DLSCA differs from TA [5] for not requiring extensive prior knowledge. Instead, deep learning models automatically learn the relationship between physical leakages (traces) and labels, classifying traces through a neural network that maps them to labels based on the leakage patterns of sensitive operations.

During training, the neural network learns a classification function $\mathcal{F}_\theta$ that maps input traces $\mathbf{T}_i$ to score vectors $\vec{p} = \mathcal{F}_\theta(\mathbf{T}_i)$, where S is the number of possible classes. The network parameters θ are updated via backpropagation and an optimizer to minimize the batch loss:

$$\theta_{\text{epoch}} = \theta_{\text{epoch-1}} - \alpha \nabla_\theta L_{\text{batch}}, \quad (1)$$

where α is the learning rate. In the attack phase, the trained model classifies each attack trace and generates score vectors for key guessing. For AES in specific, in the first and last rounds of the en-/decryption, inference of the the neural network can be performed byte-by-byte (8 bits, yielding 256 possible values). This makes DLSCA particularly effective in attacks this cryptographic primitive.

¹<https://github.com/hfl110529/CA-SCA>

B. Unsupervised Domain Adaptation

Unsupervised Domain Adaptation (UDA) [28] is a key technique in deep learning for improving model performance in unfamiliar domains, particularly when labeled data is scarce. The core idea is to leverage unlabeled data, enabling the model to learn general features across domains or modes, thereby enhancing robustness and adaptability to unseen data.

A common UDA method is Maximum Mean Discrepancy (MMD) [23], [29], which measures feature distribution differences between the source (X_s) and target (X_t) domains as:

$$\text{MMD}(X_s, X_t) = \|\mathbb{E}_{X_s}[\phi(X_s)] - \mathbb{E}_{X_t}[\phi(X_t)]\|^2, \quad (2)$$

where $\phi(\cdot)$ maps inputs to a high-dimensional space. By minimizing the combined loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{classification}} + \lambda \cdot \mathcal{L}_{\text{MMD}}(X_s, X_t), \quad (3)$$

we jointly optimize the classification loss ($\mathcal{L}_{\text{classification}}$) and domain alignment loss, enhancing the model's generalization in the target domain. Here, λ is a weighting parameter that balances the MMD loss and classification loss.

C. Evaluation metric

Guessing Entropy (GE) [30] is a common security metric in SCA. Given N_{attack} traces in the attack phase, the attacker estimates a key guessing vector $\mathbf{k}_{\text{guess}} = [k_1, k_2, \dots, k_{|\mathcal{K}|}]$, where $|\mathcal{K}|$ denotes the size of the keyspace, arranged in decreasing order of likelihood. When the rank of the true key k^* reaches zero in $\mathbf{k}_{\text{guess}}$, the number of traces used is referred to as $\bar{N}_{\text{GE}}$.

III. METHODOLOGY

Recall that CA-SCA is designed based on the combination of cross-attention mechanisms and UDA to enhance network portability. The core idea is to enhance the network's generalization across different data domains by leveraging attention mechanisms to emphasize relevant features, while domain adaptation techniques align analysis and target data in a high-dimensional space. In Subsection III-A outlines the threat model for the proposed approach, Subsection III-B discusses the rationale behind the attention mechanism, and Subsection III-C details the implementation of CA-SCA.

A. Threat Model

We consider a standard profiling side-channel setting focused on power and electromagnetic (EM) SCAs to recover secret keys from encryption algorithms. This threat model is widely recognized and realistic, as numerous certification laboratories evaluate security-critical products under similar conditions daily. Power and EM side-channels are commonly exploited to compromise modern communication devices [31] and have also been used for program flow tracking [32].

In the profiling phase, we assume the attacker has access to a cloned device running a cryptographic algorithm, though its encryption mode may differ from that of the target device. Typically deployed on an embedded platform, this cloned device can be queried with known or chosen parameters (e.g.,

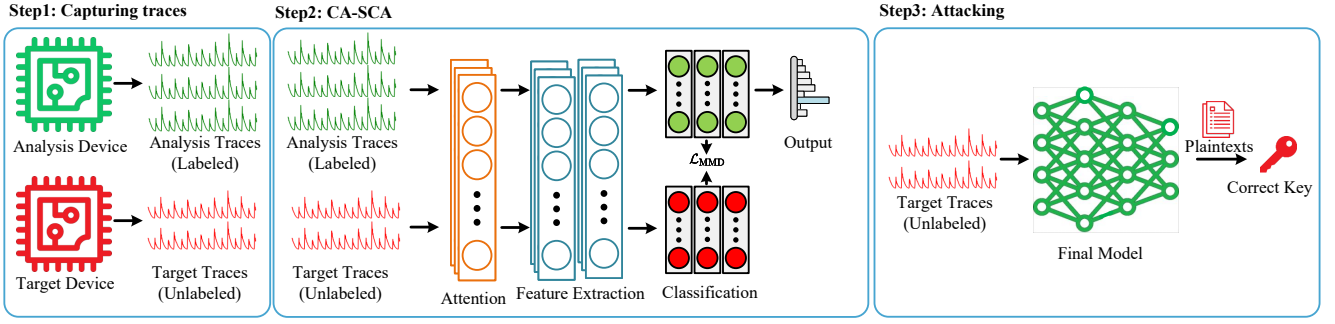


Fig. 1: Overall architecture of CA-SCA.

key, plaintext), allowing the attacker to record the resulting physical leakage (e.g., power or EM). Additionally, the attacker can acquire a limited amount of unlabeled leakage data from the target encryption mode, which is incorporated during model training to facilitate domain adaptation and address distributional differences between encryption modes.

In the attack phase, the attacker applies the trained leakage model and combines it with side-channel leakage traces from the target device to recover the encryption key. The domain adaptation technique allows the model to adapt to different encryption modes, enabling effective SCA even without labeled data from the target mode during training.

B. Attention Layer

Given an input sequence $X = [x_0, \dots, x_{n-1}] \in \mathbb{R}^{n \times d}$, the attention layer computes an output sequence $\hat{V} = [\hat{v}_0, \dots, \hat{v}_{n-1}] \in \mathbb{R}^{n \times d_v}$ as follows:

$$\hat{v}_i = \sum_{j=0}^{n-1} \text{softmax} \left(\frac{q_i^T k_j}{\sqrt{d}} \right) v_j = \sum_{j=0}^{n-1} \frac{\exp \left(\frac{q_i^T k_j}{\sqrt{d}} \right)}{\sum_{l=0}^{n-1} \exp \left(\frac{q_i^T k_l}{\sqrt{d}} \right)} v_j, \quad (4)$$

where

$$q_i = W_q x_i, \quad k_i = W_k x_i, \quad v_i = W_v x_i,$$

for $i = 0, \dots, n-1$, with $W_q, W_k \in \mathbb{R}^{d_k \times d}$ and $W_v \in \mathbb{R}^{d_v \times d}$. The final output of the attention layer is given by:

$$\hat{Y} = \hat{V} W_o^T + X,$$

where $W_o \in \mathbb{R}^{d \times d_v}$ is the projection matrix.

Multi-Head Attention: A single attention operation using parameters W_q, W_k, W_v is referred to as an attention head. In practice, multiple attention heads are employed in parallel, forming an H -head attention layer as follows:

$$\begin{aligned} \hat{V}^{(i)} &= f_A(X; W_q^{(i)}, W_k^{(i)}, W_v^{(i)}), \quad i = 0, \dots, H-1, \\ \hat{V} &= \text{concat}(\hat{V}^{(0)}, \dots, \hat{V}^{(H-1)}), \\ \hat{Y} &= \hat{V} W_o^T + X, \end{aligned} \quad (5)$$

where concat denotes row-wise concatenation of the output matrices. This enables multi-head attention to capture information from different subspaces.

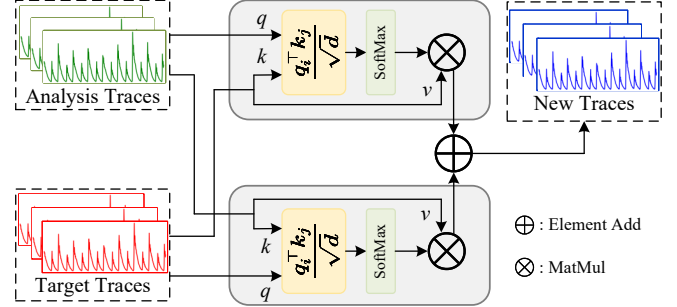


Fig. 2: Flowchart of the cross-attention mechanism.

Algorithm 1 CA-SCA Optimization Procedure

- 1: **Input:** Labeled analysis dataset $X_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$, Unlabeled target dataset $X_t = \{x_i^t\}_{i=1}^{n_t}$
- 2: **Output:** Optimized CNN model M_{best}
- 3: Split X_s into training set $X_{s,\text{train}}$ and validation set $X_{s,\text{val}}$
- 4: Initialize CNN model M with attention layer, feature extraction layer, and classifier parameters
- 5: Set initial best validation loss $val_loss_{\text{best}} \leftarrow \infty$
- 6: **while** not M .convergence **do**
- 7: Forward pass on $X_{s,\text{train}}$ and X_t using M
- 8: Compute total loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{classification}} + \lambda \cdot \mathcal{L}_{\text{MMD}}$
- 9: Update M parameters via backpropagation to minimize $\mathcal{L}_{\text{total}}$
- 10: Evaluate $val_loss = \text{Evaluate}(M, X_{s,\text{val}}, X_t)$
- 11: **if** $val_loss < val_loss_{\text{best}}$ **then**
- 12: $val_loss_{\text{best}} \leftarrow val_loss$
- 13: $M_{\text{best}} \leftarrow M$
- 14: **end if**
- 15: **end while**
- 16: **return** M_{best}

C. Tailoring CA-SCA for SCA

To implement UDA with an attention layer, we modified the model architecture to simultaneously optimize both classification loss and MMD loss (as shown in Fig. 1). This architecture enhances feature representation of both analysis and target data through an interactive attention mechanism, facilitating effective generalized learning.

The training process begins with a classic CNN stage, where

the parameters θ of the classifier f_θ are initialized, and the model parameters are optimized by minimizing the cross-entropy loss $\mathcal{L}_{\text{classification}}$ on the analysis dataset:

$$\mathcal{L}_{\text{classification}}(f_\theta) = \sum_{x_s^i, y_s^i \in \mathcal{X}_s} l(f_\theta(x), y). \quad (6)$$

Next, the target data undergoes feature extraction and, together with the analysis data, is fed into the attention layer. The attention layer computes a correlation matrix between the data, guiding the network to focus on key features, as shown in Fig. 2.

Throughout the training process (Algorithm 1), the total loss function $\mathcal{L}_{\text{total}}$ combines classification and MMD losses: the classification loss is calculated using labeled analysis data, while the MMD loss is based on features from both analysis and target data. Both losses jointly update all trainable model parameters, promoting effective generalization to the target data:

$$\theta' \leftarrow \theta - \alpha \frac{\partial \mathcal{L}_{\text{total}}}{\partial \theta}, \quad (7)$$

where α is the learning rate.

IV. EXPERIMENTS

A. Dataset

AES supports multiple encryption modes—ECB, CBC, CTR, CFB, and OFB—each influencing data processing and security differently. Although based on the same core algorithm, operational differences result in varying resistance to SCA, highlighting the need for cross-mode evaluation in comprehensive security analysis.

To evaluate generalization across modes, traces were collected from the CW308T-STM32F3 target board for each AES mode. Following the methodology in [33], Points of Interest (POIs) from the final encryption round were selected due to their favorable trace characteristics. Each dataset comprises 25,000 traces with 500 sampling points, and specific leakage models capturing key encryption operations are summarized in Table I.

To further analyze mode differences, we applied Normalized Inter-Class Variance (NICV [34]) to detect physical leakage variations. As shown in Fig. 3(a), leakage quality and timing vary across modes (e.g., CTR mode shows a time shift). Maximum Mean Discrepancy (MMD) was also used to quantify numerical differences, as shown in Fig. 3(b). Diagonal values indicate same-mode comparisons with minimal MMD values, while cross-mode comparisons yield higher values. Notably, CFB and OFB exhibit smaller MMD values due to structural similarities, while larger values highlight greater challenges in achieving generalization across different modes.

B. Setups

In the profiling phase, 20,000 traces are used, with 80% allocated for training and 20% for validation, while 500 unlabeled attack traces from the target dataset are reserved for model constraints. During the attack phase, 5,000 attack

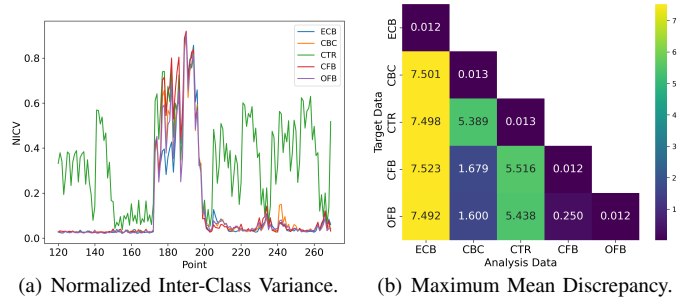


Fig. 3: Analysis of AES mode variations using NICV and MMD metrics.

TABLE I: Leakage modeling for various encryption modes, where p_t denotes the plaintext, c_t denotes the ciphertext, SBox^{-1} represents the inverse SBox operation, InvSR denotes the inverse shift rows operation, and k_{R10}^* denotes the expanded key for round 10. All models utilize the ID leakage model.

Leakage model	Encryption mode
$\text{InvSR}(c_t) \oplus (\text{SBox}^{-1}[c_t \oplus k_{R10}^*])$	ECB
$\text{InvSR}(c_t) \oplus (\text{SBox}^{-1}[c_t \oplus k_{R10}^*])$	CBC
$\text{InvSR}(c_t) \oplus ((\text{SBox}^{-1}[(c_t \oplus p_t) \oplus k_{R10}^*]))$	CFB
$\text{InvSR}(c_t \oplus p_t) \oplus ((\text{SBox}^{-1}[(c_t \oplus p_t) \oplus k_{R10}^*]))$	OFB
$\text{InvSR}(c_t \oplus p_t) \oplus ((\text{SBox}^{-1}[(c_t \oplus p_t) \oplus k_{R10}^*]))$	CTR

traces are used for evaluation. The metric $\bar{N}_{\text{GE}}$ is defined as the number of attack traces required for GE to reach rank 0. To estimate $\bar{N}_{\text{GE}}$, 1,000 attack traces are randomly selected, tested 200 times, and the average result is reported.

Our focus is on assessing CA-SCA performance across different AES modes rather than fine-tuning network architecture. Accordingly, we use a simple CNN architecture with three convolutional layers and three fully connected layers. The attention layer has 8 heads with an embedding dimension of 64. The convolutional layers have filter sizes of 16, 32, and 64, and kernel sizes of 1, 50, and 3, respectively. Each fully connected layer consists of 20 neurons. *SELU* is used as the activation function for the input and hidden layers, and *softmax* for the output layer. We set the batch size to 100, epochs to 100, and the hyperparameter λ to 0.02, using the *Adam* optimizer with a learning rate of 0.001.

All experiments are conducted with PyTorch 1.8.0 and Python 3.8.0 on hardware with an Intel® Xeon® Platinum 8358P CPU @ 2.60 GHz and an NVIDIA GeForce GTX 4090 24GB GPU.

C. Results

To facilitate comparative analysis, we first evaluated the performance of DL-PA [10] across different AES modes. The attack results are illustrated in Fig. 4(a), where the x -axis represents the analysis dataset and the y -axis represents the target dataset. The CNN model is trained on the x -axis dataset and attempts to recover the key for the target dataset. As shown, the same mode (along the diagonal) can recover the

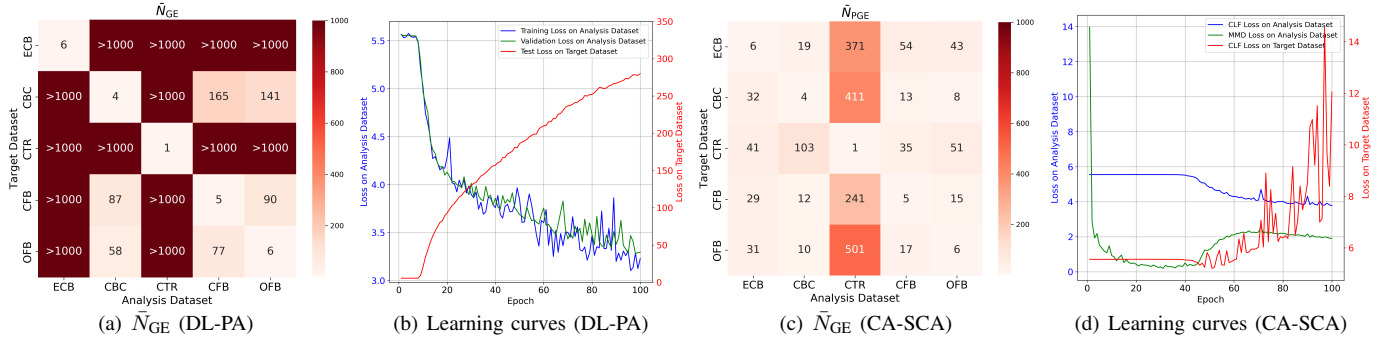


Fig. 4: Analysis results across different AES modes.

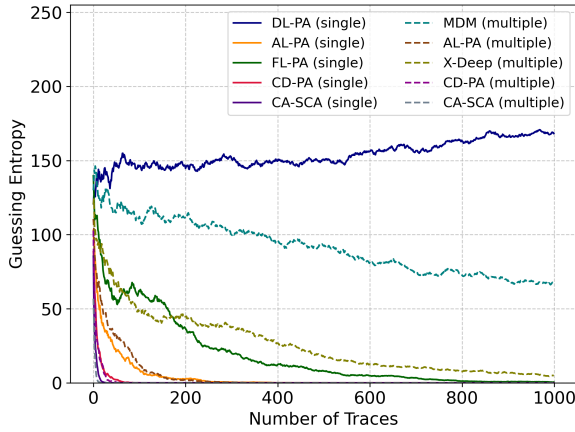


Fig. 5: Comparison with other methods. For the single-analysis dataset, training was performed on the ECB dataset and testing was performed on the OFB dataset. For multi-analysis datasets, training was performed on the ECB and CTR datasets and testing on the OFB dataset.

correct key within 10 attack traces. However, variations in AES modes may lead to reduced DL-PA performance or even failure. For instance, when the model is trained on analysis data in ECB mode, it fails to recover the key for the target mode, even after 1,000 attack traces.

To better characterize DL-PA's performance across AES modes, we present the learning curves for ECB and CBC modes in Fig. 4(b). Both training and validation losses decrease over time and stabilize around 3.2, while the testing loss on the target dataset continues to increase. This suggests that the discrepancy between target and analysis datasets is too large, leading to model failure through misclassification. In DL contexts, this is typically attributed to overfitting. However, in practical SCA, overfitting is difficult to detect without access to labeled attack traces, which prevents calculating test losses.

Fig. 4(c) summarizes CA-SCA's performance across AES modes, showcasing improved results in all cases, with some attacks nearly matching the performance achieved in identical mode scenarios. Notably, when CTR mode is used as the analysis dataset, its data-independent nature results in

more pronounced leakage features. While the CNN model effectively learns these features, it struggles to generalize to other AES modes. The learning curve of CA-SCA for the ECB-CBC task, shown in Fig. 4(d), reveals that during the initial 40 epochs, the MMD loss decreases while the classification loss remains stable. Subsequently, the training loss decreases and stabilizes, but the testing loss initially decreases before rising again. This behavior suggests that the model first aligns the analysis and target datasets in high-dimensional space through attention mechanisms and MMD loss before proceeding with classification. Despite substantial differences between the datasets leading to overfitting on the analysis dataset, the intermediate model continues to perform effectively on the target data. Furthermore, as illustrated in Fig. 4(d), the results surpass those in Fig. 4(b), highlighting CA-SCA's exceptional capability to handle data variations and improve generalization across different modes.

We then compare CA-SCA with other state-of-the-art techniques. MDM [21] and X-Deep [19] are widely used for addressing portability issues, with model robustness generally increasing with the number of analytical datasets. However, MDM and X-Deep have limitations when there are substantial differences between target and analysis datasets. For example, as shown in Fig. 3(a) and Fig. 3(b), CBC, CFB, and OFB datasets are relatively similar, but differ more significantly from other datasets. To test this, we trained the model on ECB and CTR datasets and evaluated it on the OFB dataset. In Fig. 5, both MDM and X-Deep fail to recover the correct key for the target dataset. Unlike methods like AP-PA [22], FL-PA [35], and CD-PA [23], CA-SCA requires no fine-tuning or extensive prior knowledge for SCA parameter tuning. Table II provides a detailed performance comparison of the tested methods. In summary, CA-SCA requires no specific tuning knowledge, does not rely on multiple analysis datasets, and outperforms current state-of-the-art methods in certain cases.

V. DISCUSSIONS

A. Impact of Parameter λ

The parameter λ in Equation 3 is critical in Unsupervised Domain Adaptation (UDA). If λ is set too small, the network focuses primarily on optimizing $\mathcal{L}_{\text{classification}}$, reducing the

TABLE II: Detailed performance compared to related works.

Scenario	Method	Fine-tune	$\bar{N}_{GE}$
Single Profiling Dataset	DL-PA [10]	✗	-
	AL-PA [22]	✓	493
	FL-PA [35]	✗	-
	CD-PA [23]	✓	110
	This work	✗	31
Multiple Profiling Dataset	MDM [21]	✗	-
	AL-PA [22]	✓	361
	X-Deep [19]	✗	-
	CD-PA [23]	✓	74
	This work	✗	21

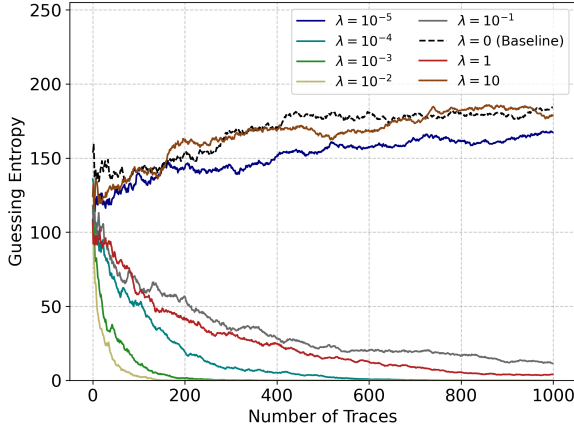


Fig. 6: Attack results of different λ on the ECB-CBC task, with $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$.

influence of $\mathcal{L}_{MMD}$ on the learning process. Conversely, setting λ too large may excessively impact the learning process, potentially leading to classification failure. To assess the sensitivity of λ , we illustrate its effect on $\bar{N}_{GE}$ in Fig. 6.

Experimental results indicate that setting λ too small (e.g., $\lambda = 10^{-5}$) or too large (e.g., $\lambda = 10$) fails to enhance attack performance, consistent with our analysis that $\lambda = 10^{-3}$ is an effective choice. While various values of λ improve performance, a practical strategy is to start with smaller values, particularly in scenarios with low signal-to-noise ratios, as this encourages the network to focus on challenging classification tasks. If the reduction in $\mathcal{L}_{MMD}$ is insufficient or progresses too slowly, λ can be incrementally increased to accelerate the UDA process.

B. Impact of the Number of Attention Heads

In prior experiments, we observed that setting the number of attention heads, denoted by H , to the default value of 8 provided strong performance. To evaluate CA-SCA’s sensitivity to different H values, we tested the ECB-CBC task with varying H while keeping other hyperparameters constant. For each H value, three independent networks were trained, and the $\bar{N}_{GE}$ results are summarized in Table III.

As shown in Table III, CA-SCA demonstrates notable instability for smaller H values (e.g., 2 and 4). Specifically,

TABLE III: CA-SCA attack results for different values of H on the ECB-CBC task, with T_1 , T_2 , and T_3 representing the $\bar{N}_{GE}$ results from three independent tests.

H	ECB-CBC		
	T_1	T_2	T_3
2	984	>1K	580
4	282	133	184
8	37	30	29
16	41	33	47
32	47	52	44

CA-SCA required more than 580 and 133 attack traces, respectively, to reach a $\bar{N}_{GE}$ of 0, whereas $H = 8$ achieved this with fewer than 40 attack traces across all trials. This instability for smaller H values is likely due to limited information flow to distant sample points. In contrast, CA-SCA performs well with larger H values, such as 8 and 16, though performance slightly declines at $H = 32$, possibly due to overfitting caused by the increased parameter count in the attention layer. These findings suggest an optimal range for H in CA-SCA, with values between 8 and 16 consistently delivering robust and reliable performance across the tests.

VI. CONCLUSIONS AND FUTURE WORK

In this paper, we introduce CA-SCA, a novel approach to address portability challenges across different AES modes. CA-SCA leverages cross-attention mechanisms and UDA to align input samples within a high-dimensional feature space, effectively reducing interference from encryption mode mismatches. Notably, CA-SCA does not require specific fine-tuning or multiple analysis datasets. Our evaluation on five distinct AES mode datasets shows that, while traditional SCA methods often struggle with substantial mode differences, CA-SCA enables previously infeasible attacks and outperforms state-of-the-art techniques.

In future work, we aim to further explore the impact of cross-attention mechanisms and UDA on SCA. Additionally, we plan to assess CA-SCA’s performance across various hardware platforms to advance portability research within the SCA community.

VII. ACKNOWLEDGEMENTS

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